Terna Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Department: Computer Engineering

Internal Assessment Test (IAT-1)

Class/Sem: 3 (Div-B) Date-____/____/____

Max. Marks: 20 Subject- Logical Reasoning Duration: 1hr

Academic Year: SH 2026-27

Q.no.	Questions	Marks	CO
Q1	**Each question of two marks (solve any 3)**	6	
1(a)	In a certain code language, if PENCIL is coded as QFODJM, how is MARKER coded in that same language?	2	CO1
1(b)	Find the missing number in the given number series: 4, 9, 19, 39, 79, ?	2	CO1
1(c)	If the letters of the word 'COMPUTER' are rearranged alphabetically, which letter will be fourth from the right end?	2	CO1
1(d)	Determine the next term in the alphanumeric series: B2C, E5F, H8I, K11L, ?	2	CO1
Q2	**Solve any one**	7	
2(a)	Pointing to a photograph of a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man? Analyze the family tree steps and justify your answer with a detailed blood relation breakdown.	4	CO2
2(b)	Six persons - A, B, C, D, E, and F - are sitting in a straight line facing North. C is sitting immediately to the left of B. D is sitting at one of the extreme ends and has E as his immediate neighbor. F is sitting third to the left of C. Determine the exact seating arrangement from left to right and explain the step-by-step deduction process.	3	CO2
	OR		
(a)	In a family of eight members across three generations, P is the father of Q and R. S is the granddaughter of T, who is married to P. V is the brother of S. Q is unmarried. U is the mother-in-law of R. Construct the complete family tree and determine the relationship between V and P.	4	CO2
(b)	Five friends - P, Q, R, S, and T - are sitting around a circular table facing the center. R is seated second to the right of P. Q is the immediate neighbor of neither P nor R. T is sitting to the	3	CO2

Q.no.	Questions	Marks	CO
	immediate left of P. Who is sitting to the immediate right of Q? Detail the circular arrangement logic.		
Q3	**Solve any one**	**7**	
3(a)	Five project tasks - T1, T2, T3, T4, and T5 - are to be scheduled from Monday to Friday, exactly one task per day. T3 must be scheduled before T1. T2 cannot be scheduled on Monday or Friday. T4 must be scheduled immediately after T2. Analyze the valid scheduling constraints and determine the exact day-wise schedule for all five tasks.	4	CO3
3(b)	Four persons - A, B, C, and D - belong to four different professions: Engineer, Doctor, Architect, and Teacher, not necessarily in that order. They live in four different cities: Mumbai, Pune, Nashik, and Nagpur. C is neither an Architect nor lives in Nagpur. The Doctor lives in Pune. A is an Engineer and does not live in Nagpur or Nashik. B lives in Mumbai. Match each person with their correct profession and city through analytical elimination.	3	CO3
	OR		
(a)	Four teams - Alpha, Beta, Gamma, and Delta - participate in a coding competition and secure ranks from 1st to 4th. Gamma did not secure the 1st rank. Beta secured a rank immediately better than Delta. Alpha finished behind Gamma. Analyze the relative rankings and deduce the final position of each team with complete logical justification.	4	CO3
(b)	Five colleagues - P, Q, R, S, and T - are grouped into two distinct project teams of 2 and 3 members respectively, based on their programming language proficiencies (Python and Java). P and Q are proficient in Python, while R, S, and T are proficient in Java. If P and R refuse to work in the same team, and each team must have at least one Python expert, analyze all valid grouping distributions.	3	CO3